

Hahn-Train vs. CPMG: Why the First Echo Differs at Reduced Flip Angles

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1 Background

In a spin-echo experiment, a 90° excitation pulse is followed by a train of refocusing pulses separated by the echo time TE. Two common phase conventions for the refocusing pulses are:

- **Hahn-Train:** all refocusing pulses along the x -axis ($\varphi = 0$).
- **CPMG** (Carr–Purcell–Meiboom–Gill): refocusing pulses along the y -axis ($\varphi = \pi/2$), i.e. perpendicular to the initial magnetisation after the 90_x° excitation.

For a *nominal* flip angle of 180° both conventions produce identical echo amplitudes at every echo. When the flip angle is reduced (e.g. to 120° for SAR reduction), this equivalence is broken and the two sequences yield different amplitudes already at the *first* echo. This note explains why.

2 Symmetry Argument for the 180° Case

After the 90_x° pulse, equilibrium magnetisation $M_0\hat{z}$ is tipped to $+\hat{y}$:

$$\mathbf{M}(0^+) = M_0 \hat{y}. \quad (1)$$

During the first TE/2 delay, B_0 inhomogeneity causes the transverse magnetisation to dephase. A spin at resonance offset $\Delta\omega$ accumulates phase $\varphi = \Delta\omega \cdot \text{TE}/2$.

A perfect 180° refocusing pulse *conjugates* the complex transverse magnetisation, i.e. $M_+ \mapsto M_-$, irrespective of whether the pulse axis is x or y (since both axes are equivalent under the 180° rotation group of the transverse plane). After a further TE/2 of free precession all isochromats rephase perfectly, giving the echo amplitude

$$A_{\text{echo}} = M_0 e^{-\text{TE}/T_2}. \quad (2)$$

The phase axis (x or y) is **irrelevant** for a 180° pulse; the first echo amplitudes of the Hahn-Train and CPMG are therefore identical.

3 Breaking the Symmetry at 120°

For a flip angle $\alpha < 180^\circ$ the refocusing is imperfect, and the pulse axis matters because it determines *which component* of the transverse magnetisation is inverted. The effect of a pulse of flip angle α about a transverse axis at phase φ_{pulse} on the complex transverse magnetisation M_+ is

$$M'_+ = \cos^2 \frac{\alpha}{2} M_+ + \sin^2 \frac{\alpha}{2} M_- e^{2i\varphi_{\text{pulse}}} - i \sin \alpha M_z e^{i\varphi_{\text{pulse}}}. \quad (3)$$

For $\alpha = 180^\circ$ Eq. (3) reduces to $M'_+ = M_- e^{2i\varphi}$, which is perfect conjugation independent of φ_{pulse} . For $\alpha = 120^\circ$ the coefficients are

$$\cos^2 \frac{\alpha}{2} = \frac{1}{4}, \quad \sin^2 \frac{\alpha}{2} = \frac{3}{4}, \quad \sin \alpha = \frac{\sqrt{3}}{2}, \quad (4)$$

and the pulse axis enters explicitly.

3.1 Analytical Echo Amplitudes

Assume the B_0 inhomogeneity produces a Gaussian distribution of resonance offsets with zero mean and standard deviation σ_ω , so each spin accumulates phase $\varphi \sim \mathcal{N}(0, \sigma_\varphi^2)$ with $\sigma_\varphi = \sigma_\omega \cdot \text{TE}/2$. Neglecting M_z recovery during TE (valid for $T_1 \gg \text{TE}$) and assuming instantaneous pulses, the ensemble-averaged echo amplitude after the first refocusing pulse evaluates to

$$A_{\text{Hahn}}^{(1)} = \left(\frac{3}{4} - \frac{h}{4} \right) M_0 e^{-\text{TE}/T_2}, \quad (5)$$

$$A_{\text{CPMG}}^{(1)} = \left(\frac{3}{4} + \frac{h}{4} \right) M_0 e^{-\text{TE}/T_2}, \quad (6)$$

where the dephasing factor

$$h = e^{-2\sigma_\varphi^2} = e^{-\sigma_\omega^2 (\text{TE}/2)^2 / 2.4} \in [0, 1] \quad (7)$$

measures how much coherence survives the first TE/2 dephasing period.

The two limiting cases are summarised in Table 1.

Table 1: First-echo amplitude (in units of $M_0 e^{-\text{TE}/T_2}$) as a function of dephasing strength for a 120° refocusing pulse.

Dephasing regime	h	$A_{\text{Hahn}}^{(1)}$	$A_{\text{CPMG}}^{(1)}$
None ($\sigma_\varphi \rightarrow 0$)	1	1/2	1
Intermediate	$(0, 1)$	$< 3/4$	$> 3/4$
Complete ($\sigma_\varphi \rightarrow \infty$)	0	3/4	3/4

Several observations follow directly:

1. For **no dephasing** the CPMG pulse leaves $\mathbf{M} \parallel \hat{y}$ unchanged (rotation around its own direction), giving a full echo. The Hahn pulse projects $M_y \mapsto M_y \cos(120^\circ) = -M_y/2$, giving only half the amplitude.
2. For **complete dephasing** ($h \rightarrow 0$) both sequences converge to $\frac{3}{4} M_0 e^{-\text{TE}/T_2}$, the incoherent echo from an ensemble with uniformly distributed phases.
3. For **any intermediate dephasing** the instantaneous-pulse model predicts $A_{\text{CPMG}}^{(1)} \geq A_{\text{Hahn}}^{(1)}$.

4 Finite-Pulse Effects and the Simulation Result

Equations (5) and (6) assume *instantaneous* refocusing pulses. In the Bloch–McConnell simulation, all pulses have finite duration τ_p . During the pulse, spins at offset $\Delta\omega$ continue to precess around

$\hat{z}$ at rate $\Delta\omega$ in addition to the intended rotation around the pulse axis. The effective rotation axis is tilted out of the transverse plane:

$$\mathbf{B}_{\text{eff}} = B_1 \hat{n}_{\text{pulse}} + \frac{\Delta\omega}{\gamma} \hat{z}, \quad \alpha_{\text{eff}} = \gamma |\mathbf{B}_{\text{eff}}| \tau_p = \alpha_{\text{nom}} \sqrt{1 + \left(\frac{\Delta\omega}{\gamma B_1}\right)^2}. \quad (8)$$

For the **CPMG (*y*-axis) pulse**, the z -precession during the pulse mixes the y -component into the x - z plane, directly undermining the y -component preservation that gives CPMG its advantage. For the **Hahn (*x*-axis) pulse**, the z -precession interacts differently with the x -rotation, and the net effect is slightly less detrimental to the refocused echo.

As a result, for the simulation parameters used here ($B_0 = 17 \text{ T}$, $b_0^{\text{inhom}} = 0.05 \mu\text{T}$, $\text{TE} = 30 \text{ ms}$, $T_2 = 71 \text{ ms}$, $\alpha = 120^\circ$) both echoes lie close to the large-dephasing limit $\frac{3}{4} M_0 e^{-\text{TE}/T_2} \approx 0.491$, but the Hahn first echo is marginally larger:

Sequence	Echo #1 (simulated)	Deviation from $\frac{3}{4} e^{-\text{TE}/T_2}$
Hahn-Train	0.4740	−3.5 %
CPMG	0.4571	−6.9 %

The 3.7 % difference is a direct consequence of the broken symmetry at 120° combined with finite-pulse off-resonance effects.

5 Summary

- For 180° refocusing pulses the Hahn-Train and CPMG produce **identical first-echo amplitudes** by symmetry.
- For any other flip angle (here 120°), the pulse axis breaks this symmetry: the two sequences yield different first echoes.
- In the *ideal* (instantaneous-pulse) picture, CPMG gives a higher first echo because it preserves the dominant y -magnetisation. In practice, finite pulse duration and B_0 inhomogeneity during the pulse reverse this advantage slightly.
- The **clinically relevant advantage of CPMG** does not lie in the first echo but in *subsequent* echoes, where stimulated-echo coherence pathways accumulate constructively, suppressing T_2^* effects and giving a stable, monotonically decaying echo train suitable for T_2 fitting.